

How the Proof Works

Thomas Birnie

Start with one smooth, finite-energy fluid state u_0 , and follow what the Navier–Stokes equations force it to do. If smoothness could fail at a finite time T_* , what would the fluid have to look like on the way there?

Take a vortex as it stretches and thins. Incompressibility forces its core to narrow as it lengthens. If stretching stays ahead of diffusion, the fluid turns faster around a smaller radius. The kinetic energy can remain finite while the change in velocity across that radius becomes steeper and steeper.

That is the critical-height problem. At a finer scale, write

$$u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t), \quad \|u_\lambda(t)\|_{\dot{H}^s} = \lambda^{s-1/2} \|u(\lambda^2 t)\|_{\dot{H}^s}.$$

Below $s = \frac{1}{2}$, concentration makes the Sobolev norm smaller. Above $s = \frac{1}{2}$, it makes the norm larger. At $s = \frac{1}{2}$, the power of λ disappears. Concentration is no longer hidden or exaggerated by the choice of scale, while nonlinear transport and viscosity remain evenly matched. This is the critical height:

$$H(t) := \frac{1}{2} \|\Lambda^{1/2} u(t)\|_2^2, \quad \Lambda := (-\Delta)^{1/2}.$$

Repeated at finer scales, this is the Zeno cascade: the kinetic energy can stay finite while smaller and smaller regions carry steeper gradients.

In order for a singularity to form, velocity has to blow up. In order for that to happen, its acceleration would also have to blow up, which means its jerk has to blow up, and so on, and so on:

$$u, \quad \partial_t u, \quad \partial_t^2 u, \quad \dots$$

What we would expect to see, as we climb those time derivatives toward a singularity, is less spatial regularity, not more.

What the time derivatives reveal

Write out the first three responses:

$$\begin{aligned}\partial_t u &= \nu \Delta u - (u \cdot \nabla)u - \nabla p, \\ \partial_t^2 u &= \nu \Delta \partial_t u - (\partial_t u \cdot \nabla)u - (u \cdot \nabla)\partial_t u - \nabla \partial_t p, \\ \partial_t^3 u &= \nu \Delta \partial_t^2 u - (\partial_t^2 u \cdot \nabla)u - 2(\partial_t u \cdot \nabla)\partial_t u \\ &\quad - (u \cdot \nabla)\partial_t^2 u - \nabla \partial_t^2 p.\end{aligned}$$

By the third line, the nonlinear term is visibly branching. The binomial form writes the complete expansion at once:

$$\begin{aligned}\partial_t^{r+1} u + \sum_{a=0}^r \binom{r}{a} ((\partial_t^a u) \cdot \nabla) \partial_t^{r-a} u + \nabla \partial_t^r p \\ = \nu \Delta \partial_t^r u.\end{aligned}$$

Set

$$V_r := \partial_t^r u, \quad Q_r := \partial_t^r p.$$

The surprise appears when these time responses are measured in Sobolev energy. Write

$$E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2, \quad E'_s = \mathcal{P}_s - 2\nu E_{s+1},$$

where $\mathcal{P}_s$ is the nonlinear transfer at height s . The pressure pairing vanishes by incompressibility. At the critical height,

$$H' = \mathcal{P}_{1/2} - 2\nu E_{3/2},$$

and after n time derivatives,

$$\frac{d^n H}{dt^n} = (-2\nu)^n E_{n+1/2} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}.$$

The highest term is the Sobolev energy at height $n + \frac{1}{2}$. As temporal order rises, Sobolev height rises with it. That is the reversal. We followed velocity, acceleration, jerk, and everything above them expecting spatial regularity to fall away. Navier–Stokes points upward through the Sobolev scale.

Viscosity gives this relation inside every temporal response. For

$$E_{r,s} := \frac{1}{2} \|\Lambda^s V_r\|_2^2,$$

the corresponding row is

$$E'_{r,s} = \mathcal{P}_{r,s} - 2\nu E_{r,s+1}.$$

The index r tells us which time response we are following; s tells us how high it sits in spatial Sobolev regularity. The factor 2ν is the link from one spatial height to the next.

And this is why the rise matters. In three dimensions,

$$s > k + \frac{3}{2} \implies H^s(\mathbb{R}^3) \hookrightarrow C^k(\mathbb{R}^3), \quad \bigcap_{m < \infty} H^m(\mathbb{R}^3) = H^\infty(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3).$$

Higher time responses have opened higher Sobolev embeddings. If every finite height reaches T_* , then every finite spatial derivative reaches it too.

How the heights reach the endpoint

Seeing a height in the identity is only half the job. It still has to reach T_* . This is what the rectangle estimates do. Choose a finite temporal depth N and a finite spatial height M . For each $n \leq N$, the adjacent pair

$$V_n \in L_t^2 H_x^{M+1}, \quad V_{n+1} = \partial_t V_n \in L_t^2 H_x^{M-1} \implies V_n \in C_t H_x^M$$

carries the middle spatial height continuously to T_* . Stacking finitely many of these rows gives $\mathcal{R}_{N,M}[u_0]$.

The whole-terminal rectangle theorem supplies that finite block for every finite (N, M) . All of the blocks come from the fluid generated by u_0 , so their shared entries agree. They fit together in $\mathcal{I}[u_0]$, whose velocity belongs to

$$\bigcap_{r,m < \infty} H^r(0, T_*; H^m(\mathbb{R}^3)).$$

This is the intersection between the time derivatives and the spatial derivatives. It contains every finite height exposed by the calculation above.

Its zeroth temporal row is $\mathcal{S}_0 := \pi_0 \mathcal{I}[u_0]$, and its endpoint is

$$u_* := u(T_*) \in \bigcap_{m < \infty} H^m(\mathbb{R}^3) = H^\infty(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3).$$

The singularity picture called for spatial regularity to disappear as the time responses rose. The intersection reaches T_* with every finite spatial height intact.

Local smooth existence continues the fluid from u_* through $[T_*, T_* + \delta]$, and uniqueness joins that continuation to the fluid we started with. A finite maximal time can be crossed, so

$$T_* = \infty.$$

Where the estimates come from

Once the intersection contains every finite Sobolev height, choose whichever height a particular estimate asks for. Three familiar choices are

$$\begin{aligned}\mathcal{I}[u_0] &\longrightarrow C_t H_x^1 \longrightarrow L_t^\infty L_x^6, \\ \mathcal{I}[u_0] &\longrightarrow C_t H_x^2 \longrightarrow L_t^\infty L_x^\infty, \\ \mathcal{I}[u_0] &\longrightarrow C_t H_x^3 \longrightarrow L_t^\infty W_x^{1,\infty}.\end{aligned}$$

The third line, for example, gives on any finite interval $[0, T]$

$$\int_0^T \|\nabla u(t)\|_{L^\infty} dt \leq CT \sup_{0 \leq t \leq T} \|u(t)\|_{H^3} < \infty.$$

For any other finite demand, ask how many derivatives and which spatial norm it needs. Choose a Sobolev height above that demand, and the intersection supplies the estimate.

The familiar continuation estimates are read from the intersection afterward. They are derived from u_0 through the intersection, not the other way around.

The entire proof can now be read in one line:

$$\begin{aligned}u_0 &\longrightarrow \{(V_n, Q_n)\}_{n \geq 0} \longrightarrow \{\mathcal{R}_{N,M}[u_0]\}_{N,M < \infty} \longrightarrow \mathcal{I}[u_0] \\ &\longrightarrow \mathcal{S}_0 \longrightarrow u_* \in H^\infty \longrightarrow T_* = \infty.\end{aligned}$$